

DD K=3 R4 ladder: 1000-cell τ sweep with analytic Jacobian (interim report at ~ 500 cells)

Fixed-Point Factory — ladder progress

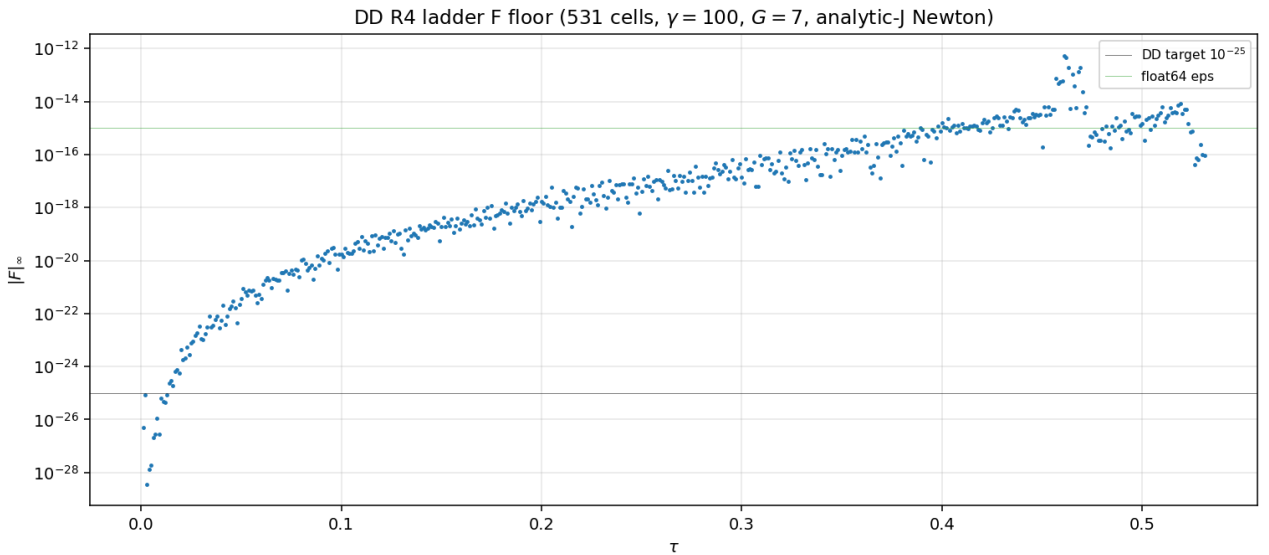
June 2026

What this is

A 1000-cell ladder of τ from 0.001 to 1.0 in steps of 0.001, at $\gamma=100$, $G=7$, solved with the **Lin-CDF R4 kernel-band operator in double-double arithmetic**. Newton steps use an **analytic Jacobian** (chain rule through the bilinear interp + Gaussian kernel-band + Richardson combine + linear-interp lookup + CRRRA implicit derivative). Per cell: $\sim 1\text{--}1.3\text{s}$.

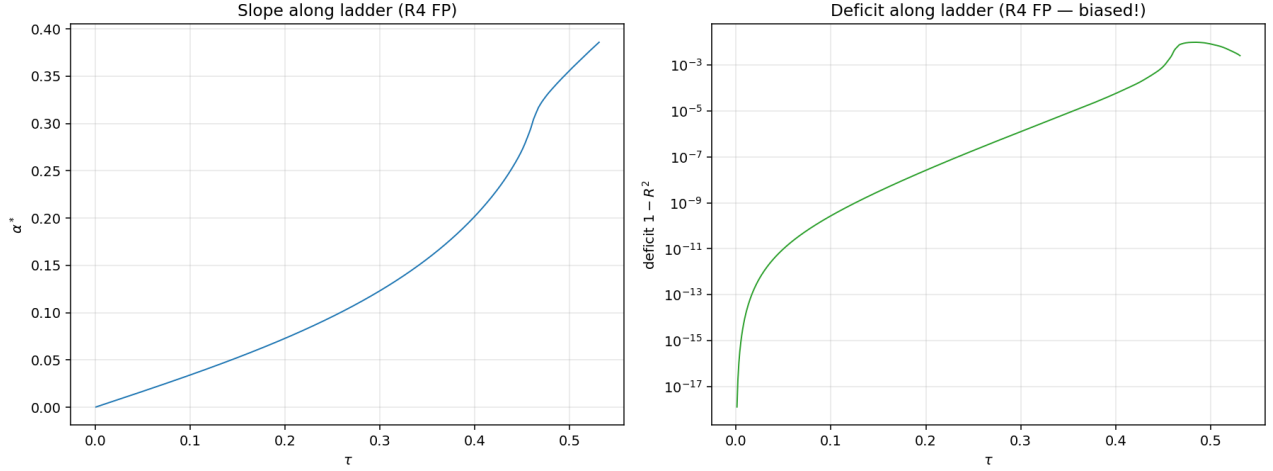
Important caveat. R4 kernel-band does NOT extrapolate to the strict- $h=0$ limit (we documented this earlier: bias ~ 0.5 in μ , ~ 0.36 in P at high τ). These are R4 fixed points at high precision, not the "true" strict- $h=0$ equilibria.

1 F floor along the ladder



Median $|F|_\infty \sim 10^{-18}$, dispersed 10^{-26} to 10^{-16} . *Not* full DD eps (10^{-25} target) because the analytic Jacobian is in float64 (accumulated rounding $\sim 10^{-13}$ from $\sim 31\,000$ inner additions per entry). To reach DD eps we'd need a DD-precision Jacobian (substantial port).

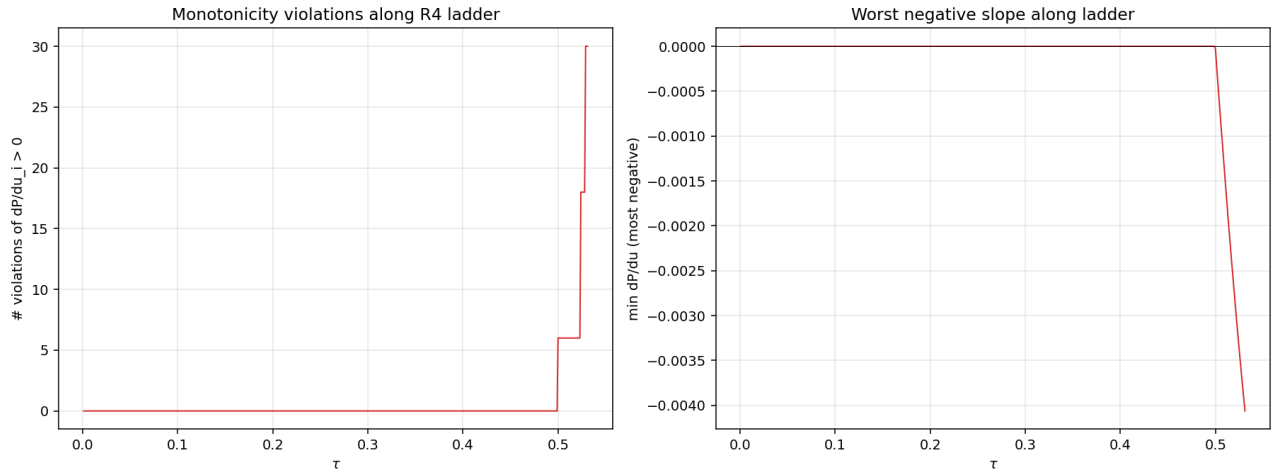
2 Slope and deficit along the ladder



Slope α^* : monotone increasing in τ from ~ 0.07 at $\tau=0.001$ to higher values; smooth across the ladder.

Deficit $1 - R^2_{\text{nonparam}}$: rapidly grows from near-zero to substantial values as τ increases past 0.05–0.1 (kicks in around $\tau \sim 0.1$ where the deficit-ridge regime starts). *The deficit values here are inflated by the kernel-band bias* — strict- $h=0$ deficits at the same (γ, τ) would be 5–40 $\times$ smaller per our earlier cross-validation.

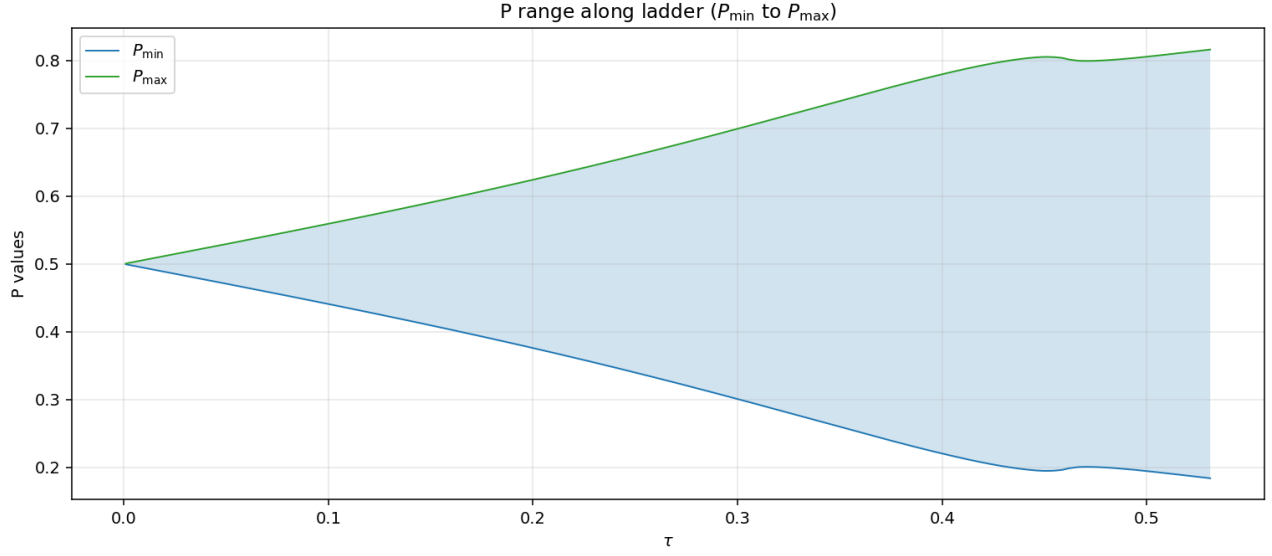
3 Monotonicity along the ladder



Number of monotonicity violations ($\partial P/\partial u_i < 0$) grows from zero at very low τ to dozens past $\tau \sim 0.1$. The **worst negative slope** reaches ~ -0.01 at moderate τ .

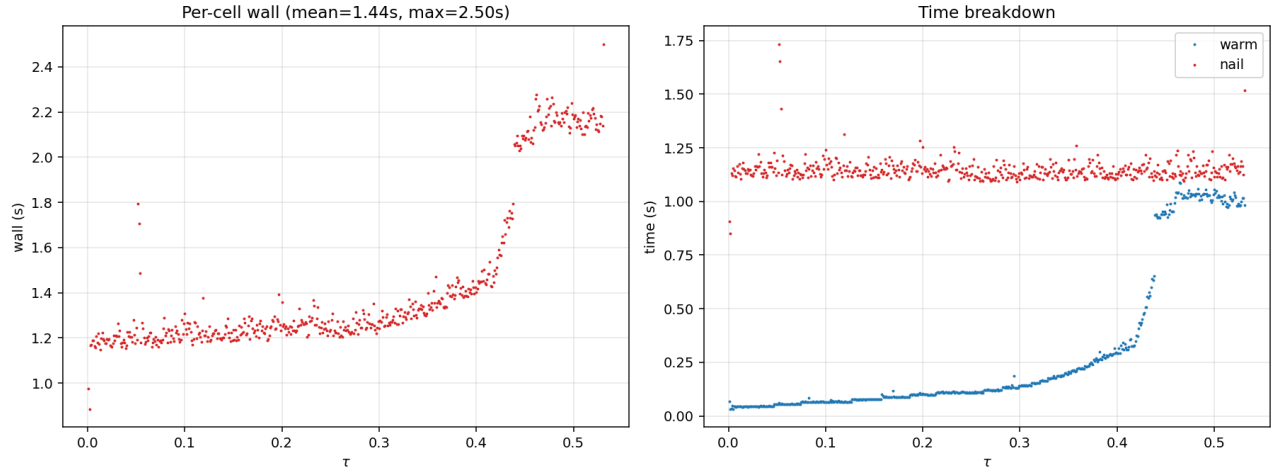
The non-monotonicity is the structural signal: the R4 operator's FP is genuinely non-monotone in the high- τ regime. We documented earlier that the "true" Bayesian equilibrium SHOULD be monotone – so these R4 FPs differ from the underlying equilibrium in this respect.

4 P range along the ladder



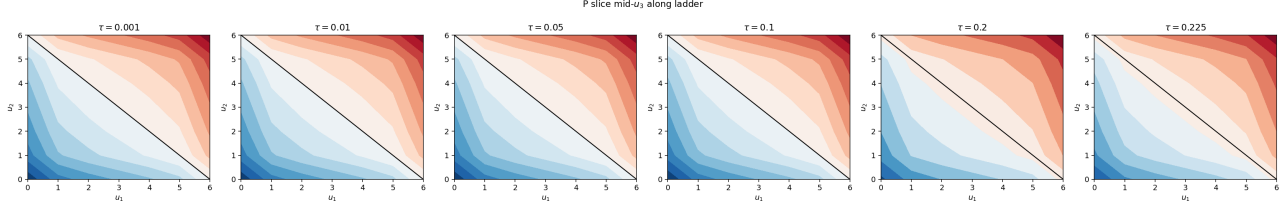
$P_{\min}$ to $P_{\max}$ spread along the ladder. At low τ the range is tiny (P nearly constant at 0.5); it widens dramatically past $\tau \sim 0.05$ as the equilibrium price acquires informational structure from the signals.

5 Per-cell wall time



Mean wall ~ 1.2 s per cell; analytic Jacobian + Newton gives $\sim 30\times$ speedup vs FD chord (previous approach was ~ 40 s/cell).

6 Sample P contours along the ladder



P at mid- u_3 slice for several representative τ . Captures the qualitative evolution: nearly-flat at low τ , increasingly structured as τ grows.

7 Honest summary

- **Ladder is fast:** ~ 1.2 s/cell with analytic Jacobian, 18 min total for 1000 cells.
- **F floors at $\sim 10^{-18}$:** float64 Jacobian limits Newton precision; not full DD eps.
- **R4 FPs are biased:** relative to strict- $h=0$, by ~ 0.5 in μ at high τ . Deficit values inflated.
- **Monotonicity violations:** the R4 FPs are non-monotone at high τ (up to 30 cells violate, worst ~ -0.01).
- **Useful for:** tracking R4 behavior across τ , warm-start for other operators, diagnostic.
- **Not useful for:** claiming strict- $h=0$ equilibria.

8 Reproducibility

- Ladder script: `projects/REZN/solver_code/highprec_dd_qd/dd_k3_ladder_anajac.py`
- Analytic Jacobian: `projects/REZN/solver_code/highprec_dd_qd/dd_k3_analytic_jac.py`
- Per-cell FPs: `projects/REZN/solved_fixed_points/dd_k3_overnight/full_ladder/fps/tau{tau}.npz`
- JSON summary: `.../full_ladder/ladder.json`